



EvenNICER-SLAM: Event-based Neural Implicit Encoding SLAM

Semester Project

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Abstract

The advancement of dense visual simultaneous localization and mapping (SLAM) has been greatly facilitated by the emergence of neural implicit representations. Neural implicit encoding SLAM, a typical example of which is NICE-SLAM, has recently demonstrated promising results in large-scale indoor scenes. However, these methods typically rely on temporally dense RGB-D image streams as input in order to function properly. When the input source does not support high frame rates or the camera movement is too fast, these methods often experience crashes or significant degradation in tracking and mapping accuracy. In this project, we propose EvenNICER-SLAM, a novel approach that addresses this issue through the incorporation of event cameras. Event cameras are a novel type of bio-inspired cameras that respond to intensity changes instead of absolute brightness. Specifically, we integrated an event loss backpropagation stream into the NICE-SLAM pipeline to enhance camera tracking with insufficient RGB-D input. We found through quantitative evaluation that EvenNICER-SLAM, with an inclusion of higher-frequency event image input, significantly outperforms NICE-SLAM with reduced RGB-D input frequency. Our results suggest the potential for event cameras to improve the robustness of dense SLAM systems against fast camera motion in real-world scenarios.

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Chapter 1

Introduction

Dense visual Simultaneous Localization and Mapping (SLAM) has been a popular subfield of 3D computer vision recently for its industrial applications in autonomous vehicle navigation, mixed/augmented reality (MR/AR), etc. While dense SLAM studies had been based on traditional methods [8, 11, 16], there has been a shift towards learning-based approaches [1, 24, 19] in recent years. Particularly, the incorporation of Neural Radiance Field (NeRF) [10] has been transformative for its capability to represent scenes with decent photometric accuracy. iMap [18] was the first real-time dense SLAM method that encodes scenes into neural implicit representations. NICE-SLAM [25], built on top of iMap, further enabled accurate camera tracking and mapping in larger indoor scenes by optimizing multi-level spatially partitioned feature grids (See Sec. 3.3).

Despite its success with spatially dense camera poses in indoor scenes, NICE-SLAM [25] is prone to failure in cases of large inter-frame camera translation or rotation, which occurs when the camera moves quickly in relation to the frame rate. Our experiments (Table 4.1, 4.2) show that while NICE-SLAM works well with a full sequence from the Replica dataset [17], it mostly crashes when the same sequence is only fed every fifth frame. To be more specific, we can do a rough estimation of average camera pose translation between adjacent frames. According to our observations, the camera travels a total distance around 20 meters throughout a Replica sequence of 2000 frames. That converts to roughly 1cm/frame, and 5cm/frame with the reduced input. In other words, NICE-SLAM would likely crash with a Replica RGB-D sequence where the camera travels an average of 5cm every frame.

In scenarios involving high-speed camera motion, however, the requirements are significantly higher. Autonomous racing drones, for example, are known to fly at maximum speeds of over 30m/s in recent years [6]. As most RGB-D cameras operate at 30 frames per second (FPS), the maximum camera pose translation between two consecutive frames would be around 1 meter. This is two order of magnitude larger than the estimated inter-frame translation for the failure cases of NICE-SLAM [25]. Additionally, there are also scenarios where cameras regularly experience high-speed rotation, such as mixed reality headsets. The Microsoft HoloLens 2, for example, is a representative device of this type of application, and according to the technical report of HoloLens 2 [21], the equipped depth camera can only operate at a maximum of 5 FPS for high resolution (1024×1024 , similar to the 1200×680 resolution in Replica [17]) depth map output. Nevertheless, Guan et al. [5] pointed out that the peak speed of human head movement can reach 240 degrees per second, which means the maximum rotation per frame would be about 48 degrees in this case. In contrast, we do not observe such large inter-frame rotations even in the reduced Replica sequence. Based on the inference above, it is clear that NICE-SLAM, as one of the recent works in the field of neural implicit encoding SLAM, is not currently applicable in high-speed scenarios.

In an effort to address the above limitation of NICE-SLAM [25], we considered the recent incorporation of event cameras [2] in computer vision research. Event cameras have high temporal resolution, high dynamic range, and low latency, making them particularly suitable for high-speed applications. Further introduction of event cameras can be found in Sec. 2.3.

In this project, we propose EvenNICER-SLAM, a novel dense SLAM system that utilizes lower-frequency RGB-D input and higher-frequency event image input. EvenNICER-SLAM is built upon the NICE-SLAM

[25] framework by adding a new event loss backpropagation stream. We conduct quantitative evaluations in camera tracking and mapping, and our results demonstrate that EvenNICER-SLAM has a large advantage in both sections over NICE-SLAM with the same reduced RGB-D streams as input. Note that we reduce the frame frequency to virtually enlarge the inter-frame camera pose translation and rotation. To the best of our knowledge, we are the first to show that the integration of event input can boost neural implicit encoding SLAM with low RGB-D frame rates in both camera tracking and scene reconstruction. We believe that EvenNICER-SLAM will unlock the potential of neural implicit encoding SLAM in high-speed applications.

Chapter 2

Related Work

In this chapter, we start from presenting multiple previous works on dense visual SLAM and NeRF (Neural Radiance Field) in the context of SLAM. Then, we proceed to offer an introduction of event cameras and event-based SLAM.

2.1 Dense Visual SLAM

Recent studies in the field of visual SLAM have primarily adopted the decoupled parallel structure of tracking and mapping, first introduced in the seminal PTAM [8]. Building upon the sparse approach of PTAM, DTAM [11] was developed to achieve dense visual SLAM in a similar decomposed manner.

An alternative approach to directly optimizing depth maps has been proposed by several studies [1, 24, 19], in which the optimization is performed on a latent scene representation that can be decoded into dense geometry. This approach alleviates the previously expensive computational cost and has subsequently led to a boost in the practicality of dense visual SLAM, particularly in terms of real-time capability. Furthermore, it can also be seen as a prototype of recent works on NeRF-driven SLAM, which are introduced later in Sec. 2.2.

2.2 NeRF in the context of SLAM

Among methods of implicit scene representation, NeRF [10] (Neural Radiance Field) has been the most extensively used one in recent years. It utilizes a large Multi-Layer Perceptron (MLP) that takes a viewing ray as input and outputs occupancy and radiance to implicitly represent scenes.

The view-dependent nature of NeRF has triggered preceding efforts to regress camera poses with an existing NeRF representation [23], which achieves a crucial prerequisite for dense visual SLAM. Additionally, a method for simultaneously fitting camera poses and a NeRF model was proposed in [9], bringing NeRF even closer to practical application in SLAM. However, the optimization of a single large MLP remained overly slow for online SLAM.

iMap [18] was the first to realize real-time dense visual SLAM using RGB-D input. However, iMap still utilizes a single MLP to represent the entire scene, which limits its performance in large scenes. NICE-SLAM [25], which is built upon iMap, addresses this limitation by encoding the scene with a hierarchical feature grid and predicting occupancy and colors with multi-level decoders. Our work is based on NICE-SLAM, and more details of its mechanism are introduced in Sec. 3.3. Furthermore, NeRF-SLAM [15] has achieved real-time dense SLAM with only RGB image sequences as input.

Despite these advances, none of the forementioned dense SLAM methods are robust against fast moving cameras and resulting motion blur. In this project, we aim to tackle this problem by integrating event input into existing NeRF-based SLAM systems.

2.3 Event Cameras and event-based SLAM

Event cameras are a bio-inspired type of camera in which each pixel responds to increases and decreases of observed logarithmic brightness asynchronously with low latency [2]. The attractive properties of event cameras, such as high temporal resolution, high dynamic range, and reduced motion blur, have made them an ideal substitute for traditional RGB cameras in various computer vision tasks involving fast camera motion.

However, real event cameras are currently still expensive and thus cannot be used extensively. To address this issue, ESIM [12, 3], an open event camera simulator, was developed to make up for the insufficient event data. As our EvenNICER-SLAM is implemented in the synthetic domain, ESIM is an essential component of our method. The usage of ESIM in our pipeline is introduced in Sec. 3.2.

Event-based SLAM is a relatively new topic that has been gaining popularity in recent years due to the benefits of event cameras mentioned above. Both Kim et al. [7] and Rebecq et al. [13] proposed real-time SLAM with only events as input. However, both methods are only able to produce semi-dense 3D point cloud as reconstruction of scenes. In contrast, our EvenNICER-SLAM makes use of neural implicit encoding to achieve dense reconstruction. More recently, Vidal et al. [22] proposed a powerful SLAM system with hybrid input of events, standard RGB images, and inertial measurements. In comparison, our method does not require an Inertial Measurement Unit (IMU) to track camera poses.

Chapter 3

Materials and Methods

In this chapter, we first give an overview of the proposed EvenNICER-SLAM framework. Then, we delve into the specific methods implemented in each module of the system. Finally, we present our optimization method using event loss.

3.1 Overview

We provide an overview of the EvenNICER-SLAM framework in Fig. 3.1. EvenNICER-SLAM is mainly based on NICE-SLAM [25], with the main difference being the reduction in the frame rate of RGB-D images fed to the SLAM, and the substitution of event images (as defined in Section 3.2) at the original frame rate.

ESIM [12, 3], an event camera simulator, is a substitute for real event cameras in the synthetic setting. We use it before running EvenNICER-SLAM to synthesize ground truth event images between all adjacent frames. Additionally, we introduce Event-Net, a differentiable event generator, to predict event images based on RGB images rendered by the SLAM. Finally, an event loss is calculated and backpropagated to optimize camera poses.

3.2 ESIM and Event-Net

Prior to conducting experiments on the SLAM system, we first synthesize event images between all adjacent frames from the original RGB image streams using ESIM [12, 3]. ESIM is an open-source event camera simulator that takes two RGB images as input and simulates events in between. We integrate the simulated events in each time interval into an event image, which records the number of positive and negative events that are fired at each pixel during a given period of time. We refer to these event images generated by ESIM as ground truth event images. Note that we let ESIM operate in the log-irradiance domain. Specifically, irradiance E (integer value ranging from 0 to 255) is converted to log-irradiance

$$L = \ln \frac{E}{255 + \epsilon}, \quad (3.1)$$

where ϵ is a parameter to assure numerical stability.

Following the generation of ground truth event images, we pretrain Event-Net to take two consecutive ground truth RGB images as input and output an event image between them. The reason for adopting a neural network to learn the behavior of ESIM [12, 3] is that our pipeline requires differentiability to backpropagate the event loss introduced in Sec. 3.4. We employ the U-Net architecture [14] for Event-Net due to its capability to capture high resolution features. We specify the numbers of input and output channels as 6 (two images, three channels each) and 2 (one for positive events and the other for negative events), respectively.

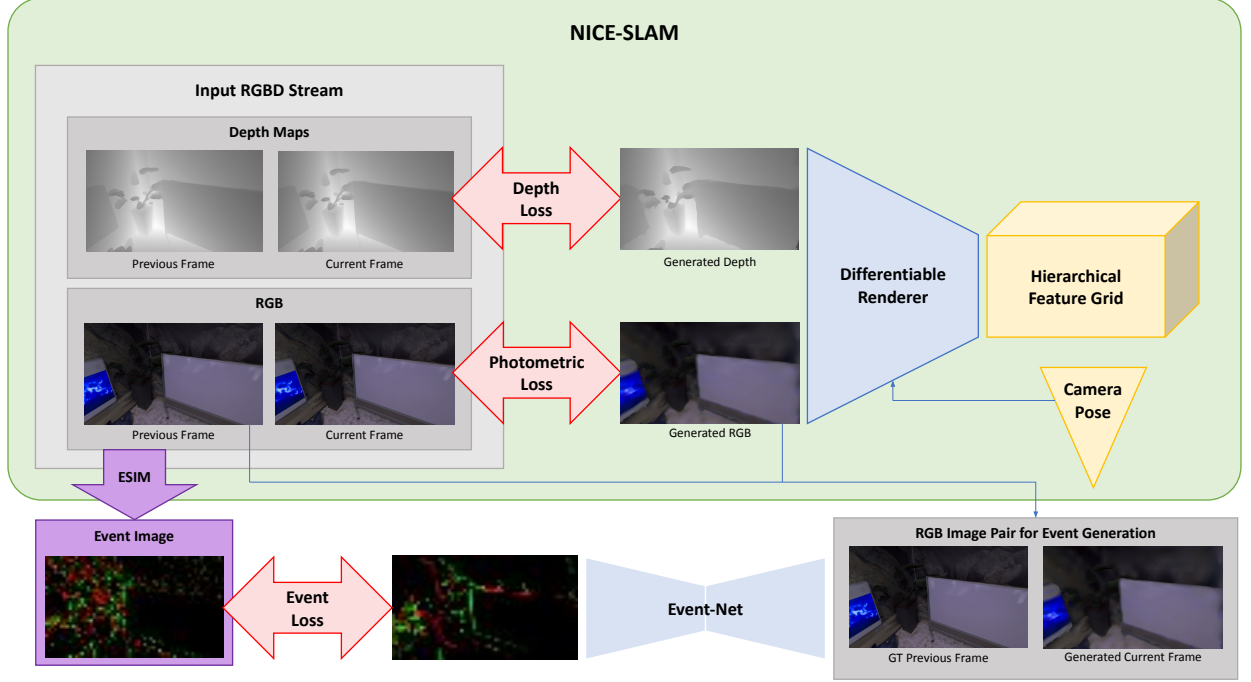


Figure 3.1: An overview of EvenNICER-SLAM framework. Before running EvenNICER-SLAM, we first generate ground truth event images between all adjacent frames in the original dataset with ESIM [12]. During a run, we feed Event-Net with the currently rendered RGB image and the latest available ground truth RGB image prior to the current timestamp, in order to predict an event image between them. Accordingly, we sum up ground truth event images at corresponding timestamps and compute an event loss. Finally, the event loss is backpropagated through the differentiable Event-Net and renderer to optimize camera poses. The components within the green box are kept the same as in NICE-SLAM [25].

3.3 NICE-SLAM

NICE-SLAM [25] takes an RGB-D stream as input and optimizes a scene representation and camera poses in an online fashion. The scene representation is implemented through a combination of hierarchical feature grid and multi-level pretrained decoders. Occupancy and color predictions from these decoders are then processed through a volume renderer to generate depth maps and RGB images. Subsequently, depth loss and photometric loss are computed and backpropagated to optimize the hierarchical feature grid and camera poses. In our work, we retain this setting as it is in NICE-SLAM.

The NICE-SLAM [25] system is parallelized into three threads: coarse-level mapper, mid-&fine-level mapper, and camera tracker. Among them, the camera tracker runs on every frame, while the mapper threads are dispatched at lower frequencies (typically every fifth frame). Currently, event loss backpropagation (Sec. 3.4) is only implemented in the camera tracker thread.

3.4 Optimization Using Event Loss

With event loss, we aim to measure the discrepancy between predicted and ground truth event images. Ideally, Event-Net should generate an event image that is similar to the corresponding ground truth from ESIM [12, 3], assuming that the SLAM predicted the transition of camera pose accurately. In practice, however, it is challenging to generate perfectly-aligned event images. Due to the high-frequency nature of event images, any pixelwise misalignment between predicted and ground truth event images can cause an irregular change of a naive pixelwise event loss. This can make the parameter space overly rough and undermine optimization. To smooth the parameter space and stabilize the optimization process, we apply a Gaussian filter to both

predicted and ground truth event images before obtaining an L_2 loss between them.

Another challenge is that rendering a full-resolution image every iteration is time-consuming and requires considerable computational resources. To address this, we only render a largely downsampled image each iteration and downscale its corresponding ground truth event image accordingly.

In the experimental setting of EvenNICER-SLAM, ground truth RGB-D images are only fed when the mapper threads run and prohibited otherwise. On the other hand, a ground truth event images is fed for every frame. To avoid error accumulation, we always use the latest available ground truth RGB image prior to the current timestamp instead of a rendered one as the first input of Event-Net. Note that it is downscaled to the same size as that of the rendered current RGB image before inference. Accordingly, we simply sum up the corresponding ground truth event images in between.

In summary, we have predicted event images:

$$\hat{I}_{event}^t = f(I_{RGB}^{t_{GT}}, \hat{I}_{RGB}^t), t - \tau \leq t_{GT} \leq t - 1, \quad (3.2)$$

where $\hat{I}_{event}$ is the predicted event image, f stands for the mapping from input to output of Event-Net, I_{RGB} and $\hat{I}_{RGB}$ are (downscaled) ground truth and rendered RGB images respectively, t is the current timestamp (frame index), τ is the period of ground truth RGB-D images being fed in relation to event images (typically $\tau = 5$), and t_{GT} is the most recent timestamp when ground truth RGB-D images are fed. Finally, we calculate the L_2 event loss:

$$L_{event} = \lambda_{event} \sum_x \sum_y \left(\text{Gaussian} \left(\sum_{t=t_{GT}+1}^t I_{event}^t(x, y) \right) - \text{Gaussian} \left(\hat{I}_{event}^t(x, y) \right) \right)^2, \quad (3.3)$$

where I_{event} is the ground truth event image, and λ_{event} is a coefficient balancing the event loss and reconstruction loss (the sum of depth loss and photometric loss).

Finally, the event loss is backpropagated through the differentiable Event-Net and renderer to optimize camera poses. We fix the parameters of Event-Net during the entire run. The learning rates are maintained the same as those of the reconstruction loss in NICE-SLAM [25].

Chapter 4

Experiments and Results

In this chapter, we first provide information on our experimental setup for reproducibility. Then, we present quantitative evaluation results for both camera tracking and mapping.

4.1 Experimental Setup

Dataset We use Replica [17] RGB-D image sequences rendered in 8 different scenes (each 2000 frames long). The sequences are directly downloaded from the NICE-SLAM [25] repository. Event images between adjacent frames are generated by ESIM [12, 3] prior to the experiments.

ESIM We fix the parameters of ESIM [12, 3] for all ground truth event images. We use positive and negative contrast threshold of $C_+ = C_- = 0.1$ and a refractory period [s] (the minimum waiting period before a pixel can trigger a new event) of $t_{ref} = 10^{-4}$. We set $\epsilon = 10^{-3}$ in Eq.3.1.

Event-Net We use an L_2 -loss between ground truth and predicted event images to pretrain Event-Net. The scale factor for downscaling is 0.1, meaning that a Replica [17] image of size 1200×680 is downscaled to 120×68 . The pretraining process is limited to 1 epoch on the forementioned Replica RGB image dataset.

Baselines We use NICE-SLAM [25] fed with RGB-D images every fifth frame (i.e. $\tau = 5$, four frames out of every five from the original RGB-D stream are excluded) as a baseline. This results in an increase in inter-frame camera translation and rotation, simulating a faster camera motion scenario. Additionally, we provide a comparison between the performance of EvenNICER-SLAM and that of a full-input NICE-SLAM.

Metrics We employ the same metrics as in NICE-SLAM [25] for our evaluation. The *ATE RMSE* [cm] is used to evaluate camera tracking error. For the evaluation of mapping quality, we consider *L1 depth loss* [cm], *accuracy* [cm], *completion* [cm], and *completion ratio* [$< 5\text{cm } \%$].

Implementation Details All experiments are run on only one NVIDIA GeForce GTX TITAN X GPU. For EvenNICER-SLAM, event images are fed for all frames except for the first, while RGB-D images are only available every fifth frame (i.e. $\tau = 5$). The tracker thread uses both RGB-D and event images when the frame ID is a multiple of 5 and only event images otherwise. The mapper threads are dispatched every fifth frame, and only use RGB-D information for mapping. For Gaussian filtering, we use a 5×5 Gaussian kernel with a standard deviation $\sigma = 1.1$ in both x and y directions. Empirically we set $\lambda_{event} = 0.025$.

4.2 Results

In Figure 4.1, we present a visual comparison of RGB images rendered under each experimental setting, providing a qualitative evaluation of the scene reconstruction. Furthermore, we present quantitative results on camera tracking and mapping in Table 4.1 and Table 4.2 respectively. These tables provide detailed numerical evaluations of the performance of each method in terms of tracking accuracy and mapping quality.

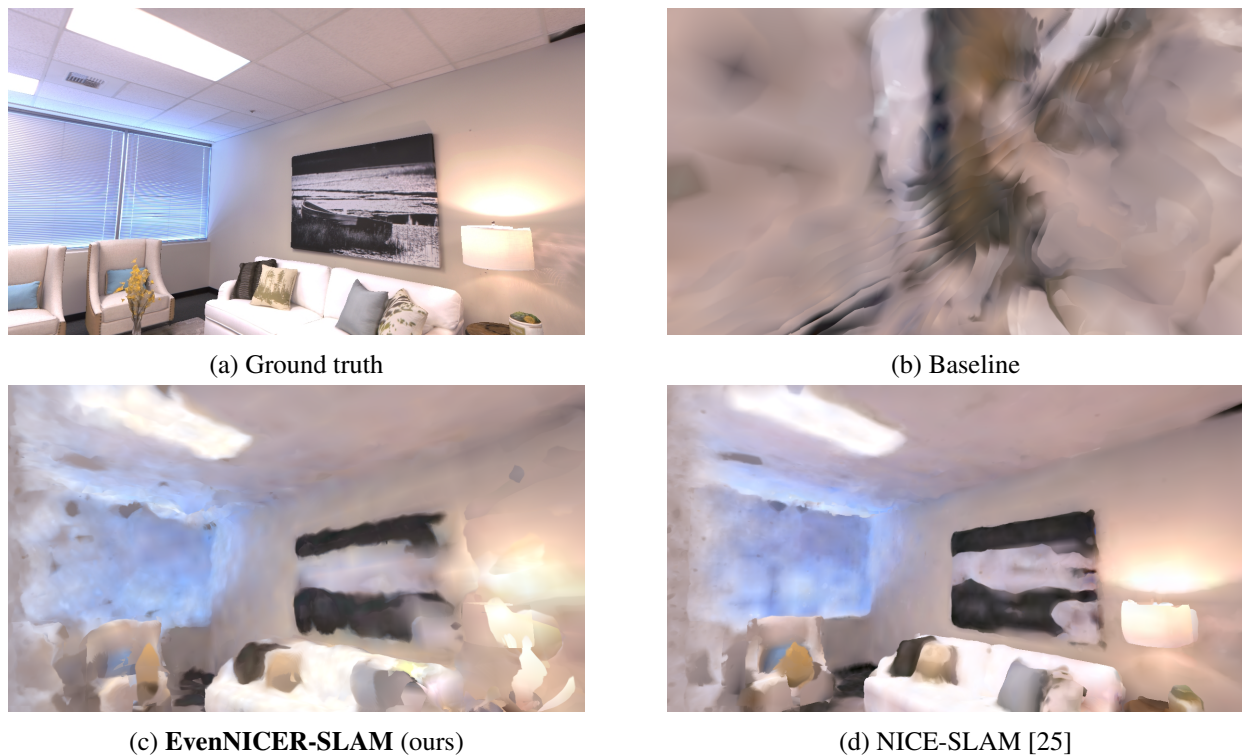


Figure 4.1: Examples of rendered images. We present a comparison between a ground truth RGB image (frame #710 from `room0`) and the corresponding images rendered by NICE-SLAM [25] with reduced frame rate (baseline), EvenNICER-SLAM, and the full-input NICE-SLAM.

Scene	room0	room1	room2	office0	office1	office2	office3	office4
Baseline	F(701)	F(380)	262.6	135.0	F(350)	F(1754)	F(658)	F(465)
Ours	43.6	89.1	19.2	42.1	67.9	50.5	42.6	F(1352)
NICE-SLAM [25]	2.5	1.9	1.5	1.3	1.2	1.7	3.4	5.5

Table 4.1: Quantitative evaluation results of camera tracking. The metric adopted here is *ATE RMSE* [cm]. An “F” indicates that the SLAM lost track of the camera and failed to complete the entire sequence. The value in parentheses following the “F” represents the number of frames out of 2000 that the SLAM system was able to process before crashing.

	Metric	room0	room1	room2	office0	office1	office2	office3	office4
Baseline	Depth L1	F	F	86.79	78.74	F	F	F	F
	Accuracy	F	F	149.92	107.19	F	F	F	F
	Completion	F	F	22.90	29.78	F	F	F	F
	Comp. Ratio	F	F	18.35	17.56	F	F	F	F
Ours	Depth L1	73.55	84.22	11.95	40.93	44.09	22.94	36.79	F
	Accuracy	50.96	71.70	8.31	43.73	37.96	13.86	25.31	F
	Completion	39.79	40.20	7.50	24.97	26.21	11.24	20.79	F
	Comp. Ratio	21.46	13.62	52.33	16.33	14.68	33.53	26.93	F
NICE-SLAM [25]	Depth L1	1.71	1.59	2.30	1.61	1.57	7.56	5.23	2.76
	Accuracy	2.69	2.24	2.27	2.42	2.52	3.40	3.40	3.06
	Completion	2.72	2.33	2.53	1.86	1.83	0.28	3.29	3.83
	Comp. Ratio	91.54	93.06	91.62	95.66	94.24	89.13	86.20	85.74

Table 4.2: Quantitative evaluation results of mapping. The metrics are *L1 depth loss* [cm], *accuracy* [cm], *completion* [cm], *completion ratio* [%]. Again, an “F” stands for a failure of the SLAM system.

Chapter 5

Discussion

In this chapter, we first provide an interpretation on our experimental results. We then discuss potential approaches to improve the results in the same problem setting. Finally, we further explore the possibility of EvenNICER-SLAM and its potential applications.

From Figure 4.1, we can see that while the original NICE-SLAM [25] gives the best reconstruction result, our EvenNICER-SLAM also produces plausible mapping. Also, it is difficult to discern the difference in camera poses from the ground truth. In contrast, the baseline SLAM loses track of the camera completely and the mapping is severely distorted. The comparison of our EvenNICER-SLAM and the baseline in Table 4.1 shows that the presence of event supervision largely reduces camera tracking error and also lowers the chance of losing track of the camera completely halfway through a 2000-frame sequence. On the other hand, there is still a non-negligible gap between EvenNICER-SLAM and full-input NICE-SLAM. We observe a similar tendency in terms of reconstruction quality from Table 4.2. Since we keep the mapping process consistent, this observation indicates a positive correlation between camera tracking accuracy and reconstruction quality.

In summary, we can first conclude that event information can positively contribute to camera tracking. Additionally, better tracking leads to better mapping. High-frequency event input can also indirectly help improve mapping quality even with lower-frequency RGB-D input. It is difficult for EvenNICER-SLAM with reduced RGB-D feed to match the performance of the original NICE-SLAM [25] as an event image essentially provides less information than a set of RGB-D images. Nevertheless, the results suggest the possibility that EvenNICER-SLAM could outperform NICE-SLAM with the same RGB-D input frequency and a higher-frequency event input.

Some approaches may help better exploit the event information and further improve the performance of EvenNICER-SLAM. One such approach is converting the event loss to a more tractable grayscale photometric loss. Specifically, we can preprocess the ground truth grayscale image by adding upon it the ground truth event image scaled by the contrast threshold. Then the resulting grayscale image can directly act as a supervision of the corresponding rendered image using a photometric loss. We expect that this approach would alleviate the alignment issue that the event loss suffer from and stabilize optimization.

Furthermore, we can also explore the use of event loss in the mapper thread. If EvenNICER-SLAM could gain meaningful information about scene geometry from event input, it would even be possible to get rid of the depth map input. This would greatly expand the range of applicable synthetic datasets and reduce the cost of data collection for potential use in real-world scenarios. One possible approach to achieving this is by first estimating dense optical flow from events, as proposed in [4]. This can then be integrated into the SLAM system, as demonstrated in the case of DROID-SLAM [20].

A more ambitious vision for EvenNICER-SLAM is to use it in the real domain. The low-latency and no-motion-blur nature of event cameras could help SLAM systems work well even with fast camera motions. A potential issue is that real event data would be much noisier than what ESIM [12, 3] simulates. To this end, adversarial training of Event-Net might be helpful in closing the domain gap between synthetic data and real data. If EvenNICER-SLAM is successfully adapted to the real domain, it could help, for example, autonomous drones and self-driving vehicles perceive their surrounding environments and plan their moves better.

Chapter 6

Conclusion

In this project, we proposed EvenNICER-SLAM, an event-based neural implicit encoding SLAM system. This is achieved by integrating synthetic event image input into the existing method NICE-SLAM [25], to keep track of camera poses with reduced RGB-D input. We quantitatively evaluated the performance of EvenNICER-SLAM and NICE-SLAM in both tracking and mapping. The comparison of their results revealed that higher-frequency feed of event images largely improves both tracking and mapping quality when RGB-D input are temporally sparse. On the other hand, the performance of EvenNICER-SLAM remains inferior to that of the original NICE-SLAM when RGB-D images are fed as frequently as event images, as event images offer less information than RGB-D images by nature. Finally, we discussed the possibility of our method to run without ground truth depth maps or on real data, and its potential future applications.

Bibliography

- [1] Michael Bloesch, Jan Czarnowski, Ronald Clark, Stefan Leutenegger, and Andrew J Davison. Codeslam—learning a compact, optimisable representation for dense visual slam. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2560–2568, 2018.
- [2] Guillermo Gallego, Tobi Delbrück, Garrick Orchard, Chiara Bartolozzi, Brian Taba, Andrea Censi, Stefan Leutenegger, Andrew J Davison, Jörg Conradt, Kostas Daniilidis, et al. Event-based vision: A survey. *IEEE transactions on pattern analysis and machine intelligence*, 44(1):154–180, 2020.
- [3] Daniel Gehrig, Mathias Gehrig, Javier Hidalgo-Carrió, and Davide Scaramuzza. Video to events: Recycling video datasets for event cameras. In *IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, June 2020.
- [4] Mathias Gehrig, Mario Millhäusler, Daniel Gehrig, and Davide Scaramuzza. E-raft: Dense optical flow from event cameras. In *2021 International Conference on 3D Vision (3DV)*, pages 197–206. IEEE, 2021.
- [5] Yongjie Guan, Xueyu Hou, Nan Wu, Bo Han, and Tao Han. Deepmix: mobility-aware, lightweight, and hybrid 3d object detection for headsets. In *Proceedings of the 20th Annual International Conference on Mobile Systems, Applications and Services*, pages 28–41, 2022.
- [6] Drew Hanover, Antonio Loquercio, Leonard Bauersfeld, Angel Romero, Robert Penicka, Yunlong Song, Giovanni Cioffi, Elia Kaufmann, and Davide Scaramuzza. Autonomous drone racing: A survey. *arXiv e-prints*, pages arXiv–2301, 2023.
- [7] Hanme Kim, Stefan Leutenegger, and Andrew J Davison. Real-time 3d reconstruction and 6-dof tracking with an event camera. In *European conference on computer vision*, pages 349–364. Springer, 2016.
- [8] Georg Klein and David Murray. Parallel tracking and mapping on a camera phone. In *2009 8th IEEE International Symposium on Mixed and Augmented Reality*, pages 83–86. IEEE, 2009.
- [9] Chen-Hsuan Lin, Wei-Chiu Ma, Antonio Torralba, and Simon Lucey. Barf: Bundle-adjusting neural radiance fields. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5741–5751, 2021.
- [10] Ben Mildenhall, Pratul P Srinivasan, Matthew Tancik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. *Communications of the ACM*, 65(1):99–106, 2021.
- [11] Richard A Newcombe, Steven J Lovegrove, and Andrew J Davison. Dtam: Dense tracking and mapping in real-time. In *2011 international conference on computer vision*, pages 2320–2327. IEEE, 2011.
- [12] Henri Rebecq, Daniel Gehrig, and Davide Scaramuzza. Esim: an open event camera simulator. In *Conference on robot learning*, pages 969–982. PMLR, 2018.
- [13] Henri Rebecq, Timo Horstschäfer, Guillermo Gallego, and Davide Scaramuzza. Evo: A geometric approach to event-based 6-dof parallel tracking and mapping in real time. *IEEE Robotics and Automation Letters*, 2(2):593–600, 2016.

BIBLIOGRAPHY

- [14] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
- [15] Antoni Rosinol, John J Leonard, and Luca Carlone. Nerf-slam: Real-time dense monocular slam with neural radiance fields. *arXiv preprint arXiv:2210.13641*, 2022.
- [16] Thomas Schops, Torsten Sattler, and Marc Pollefeys. Bad slam: Bundle adjusted direct rgb-d slam. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 134–144, 2019.
- [17] Julian Straub, Thomas Whelan, Lingni Ma, Yufan Chen, Erik Wijmans, Simon Green, Jakob J Engel, Raul Mur-Artal, Carl Ren, Shobhit Verma, et al. The replica dataset: A digital replica of indoor spaces. *arXiv preprint arXiv:1906.05797*, 2019.
- [18] Edgar Sucar, Shikun Liu, Joseph Ortiz, and Andrew J Davison. imap: Implicit mapping and positioning in real-time. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 6229–6238, 2021.
- [19] Edgar Sucar, Kentaro Wada, and Andrew Davison. Nodeslam: Neural object descriptors for multi-view shape reconstruction. In *2020 International Conference on 3D Vision (3DV)*, pages 949–958. IEEE, 2020.
- [20] Zachary Teed and Jia Deng. Droid-slam: Deep visual slam for monocular, stereo, and rgb-d cameras. *Advances in Neural Information Processing Systems*, 34:16558–16569, 2021.
- [21] Dorin Ungureanu, Federica Bogo, Silvano Galliani, Pooja Sama, Xin Duan, Casey Meekhof, Jan Stühmer, Thomas J Cashman, Bugra Tekin, Johannes L Schönberger, et al. Hololens 2 research mode as a tool for computer vision research. *arXiv preprint arXiv:2008.11239*, 2020.
- [22] Antoni Rosinol Vidal, Henri Rebecq, Timo Horstschafer, and Davide Scaramuzza. Ultimate slam? combining events, images, and imu for robust visual slam in hdr and high-speed scenarios. *IEEE Robotics and Automation Letters*, 3(2):994–1001, 2018.
- [23] Lin Yen-Chen, Pete Florence, Jonathan T Barron, Alberto Rodriguez, Phillip Isola, and Tsung-Yi Lin. inerf: Inverting neural radiance fields for pose estimation. In *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 1323–1330. IEEE, 2021.
- [24] Shuaifeng Zhi, Michael Bloesch, Stefan Leutenegger, and Andrew J Davison. Scenecode: Monocular dense semantic reconstruction using learned encoded scene representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11776–11785, 2019.
- [25] Zihan Zhu, Songyou Peng, Viktor Larsson, Weiwei Xu, Hujun Bao, Zhaopeng Cui, Martin R Oswald, and Marc Pollefeys. Nice-slam: Neural implicit scalable encoding for slam. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12786–12796, 2022.